
Which Eviction Policy Should an LLM Cache Use? A Systematic Study Across Workloads, Capacities, and Encoders

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Abstract

Semantic caches answer an incoming query with the response stored for a nearby embedding, but proposed eviction policies have not been compared under one protocol. We evaluate FIFO, LRU, LFU, ARC, GDSF, a streaming adaptation of SISO, and a semantic-redundancy policy across three ordered query corpora, three cache capacities, and two encoders. No evaluated policy improves on LFU by more than 0.0409 percentage points in any of the eighteen settings. Replacement still matters at tight capacity: FIFO and streaming SISO trail LFU by as much as 8.7 points. We explain the missing upside with a conditional packing result. Under exact lookup and insert-on-miss, a newly inserted entry cannot have a resident neighbor within the hit radius, so a geometry-aware eviction rule receives little new redundancy signal. A separate audit finds a larger problem. At a median-nearest-neighbor operating point, only 2.1–3.9% of sampled LMSYS and QQP hits are judged answer-substitutable, reducing raw hit rates of 51–60% to quality-adjusted rates of 1.1–2.2%. LFU is the strongest simple default in this protocol, but threshold calibration and hit validity require attention before sub-point eviction differences.

1 Introduction

A semantic cache embeds an LLM request and reuses a stored response when the nearest cached query falls within a distance threshold [1]. Classical policies such as LRU and LFU compete with ARC [9], GDSF [2], and policies that use embedding geometry [6, 8, 12]. Published orderings differ. Sun et al. [12], for example, report FIFO beating LRU and LFU on agent-memory workloads.

We compare seven policies while holding the query order, index, threshold, and measurement suffix fixed. The result is narrower than “eviction does not matter.” No complex policy improves meaningfully on LFU, but poorly matched policies can lose several points. We also identify when semantic redundancy is suppressed by insert-on-miss admission and test whether the nominal hits are safe response reuses.

2 Study design

Inputs. We form 100K-query corpora from the first user utterance in each LMSYS-Chat-1M conversation [13], individual Quora Question Pairs questions [5], and MOSS instructions [11]. Preprocessing removes exact duplicate text. We sample without replacement and restore source-row order. These inputs preserve corpus order but are not request traces; in particular, they contain no exact query repetition. We test capacities of 10%, 20%, and 30% under MiniLM [10] and gte-base [7].

Cache protocol. A nearest neighbor within the hit threshold is served without insertion; a miss is inserted and may trigger eviction. The cache uses FAISS HNSW ($M=32$, $efConstruction=128$, $efSearch=128$). For a capacity fraction C , we prefill the cache with the first C of the corpus, exclude the remainder of the first 30%, and measure the final 70%. Thus every capacity sees the same measured suffix, but the protocol is a capacity-sized prefill rather than a 30% online warmup. MiniLM uses $L_2^2 \leq 0.90$ (cosine ≥ 0.55 for unit vectors), its median-nearest-neighbor operating point; gte-base parameters are matched by empirical similarity quantile.

Policies and repetitions. We test FIFO, LRU, LFU, ARC, GDSF, a single-pass streaming adaptation of SISO [6], and a semantic policy that ranks redundancy relative to recency-frequency utility. Three seeds vary randomized policy-side sampling; they do not create independent traces. The maximum standard deviation over the 126 policy cells is 3.1×10^{-4} in hit-rate units.

Hit audit. At 10% capacity under MiniLM, we sample 1,000 hits per dataset-policy cell across distance quantiles, producing 13,669 unique pairs. A Llama-3.1-8B judge [4] asks whether a correct, complete answer to the cached query would also answer the incoming query. We call a YES verdict *answer-substitutable*. Rejudging 1,931 pairs with a second runtime serving the same model family gives 98.55% agreement ($\kappa = 0.826$); this checks runtime consistency, not human validity.

3 Results

3.1 No complex policy improves on LFU

Figure 1 reports the matrix. The largest gain over LFU is 0.0409 pp. ARC remains within 0.01 pp of LFU, GDSF follows LFU-with-aging, and the semantic policy adds $3.4\text{--}5.9\times$ LRU’s per-query time on LMSYS and QQP without a larger hit rate.

The weak policies show why the conclusion must stay bounded. FIFO never beats LRU and ranks sixth or seventh in seventeen of the eighteen cells; the exception is a saturated MOSS cell where all seven policies lie within 0.012 pp. Streaming SISO and FIFO differ by 0.096 pp on average across LMSYS and QQP, but streaming SISO trails LFU by up to 8.55 pp and costs $2.8\text{--}5.3\times$ LRU on those corpora. These are descriptive comparisons on one ordered corpus per dataset, not significance tests across request traces. They do not reproduce Sun et al.’s FIFO ordering [12]; differences in repetition, locality, cache semantics, or implementation remain hypotheses until tested in one shared workload.

Capacity narrows rather than eliminates every gap. LFU’s lead over LRU falls from 1.5 pp on LMSYS and 2.5 pp on QQP at 10% capacity to at most 0.012 pp at 30%; the largest streaming-SISO deficit at 30% is still 1.40 pp.

3.2 The operating point matters more than the winner

Using MiniLM’s threshold unchanged with gte-base makes every measured request a hit and produces zero evictions. After matching thresholds by similarity quantile, the leading-policy ordering returns, but LFU’s absolute hit rate moves by +19.0 pp on LMSYS and -4.6 pp on QQP. A threshold is therefore an encoder-specific operating point, and distribution matching does not establish response quality.

Figure 2 tests that quality directly. Answer-substitutable fractions are 3.1–3.9% on LMSYS and 2.1–3.1% on QQP. Their Wilson intervals overlap across policies, while quality-adjusted hit rates fall to 1.1–2.2%. Even LMSYS’s nearest distance decile, with median $L_2^2 = 0.10$, reaches only 8.9% YES. Shared prompt templates often hide different payloads. The measured threshold is useful for stressing eviction, not suitable as a deployment recommendation. Verified caching reaches a related design conclusion [3].

4 Why semantic signal disappears

Packing condition. Let C_t be the resident embeddings before query q_t , and let h be the hit radius. With exact lookup and insert-on-miss, q_t is inserted only if $\min_{c \in C_t} d(q_t, c) > h$. For any redundancy radius $r \leq h$, the inserted entry therefore has no redundancy edge to any resident entry, and every later insertion satisfies the same condition, so two stream-inserted entries are never within r of each

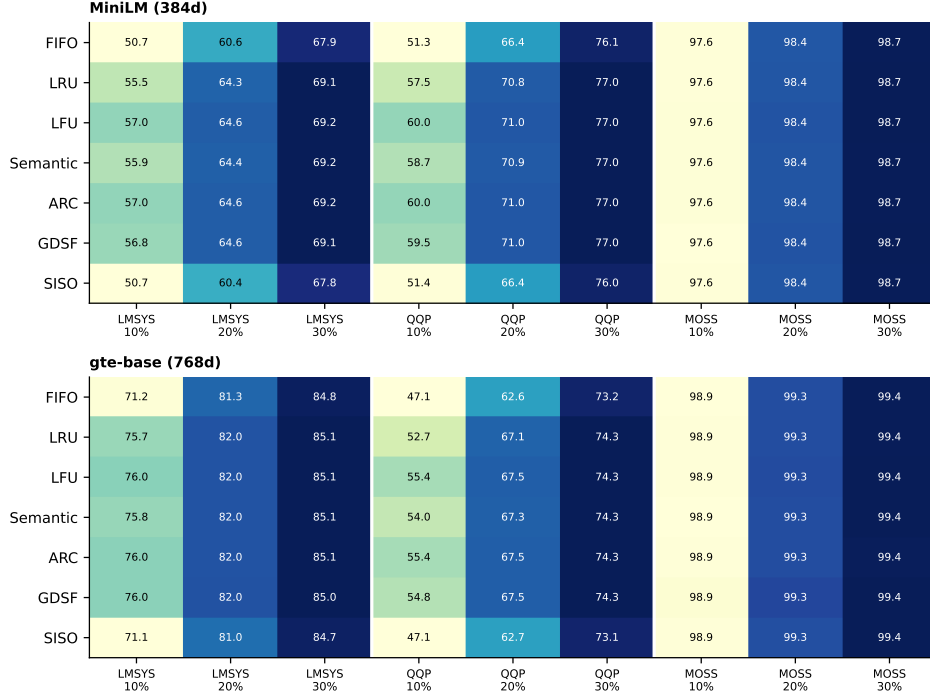


Figure 1: Hit rate (%) by policy, corpus, capacity, and encoder; color is normalized within each corpus block. No policy improves on LFU by more than 0.0409 pp, while FIFO and streaming SISO lose several points at tight capacity.

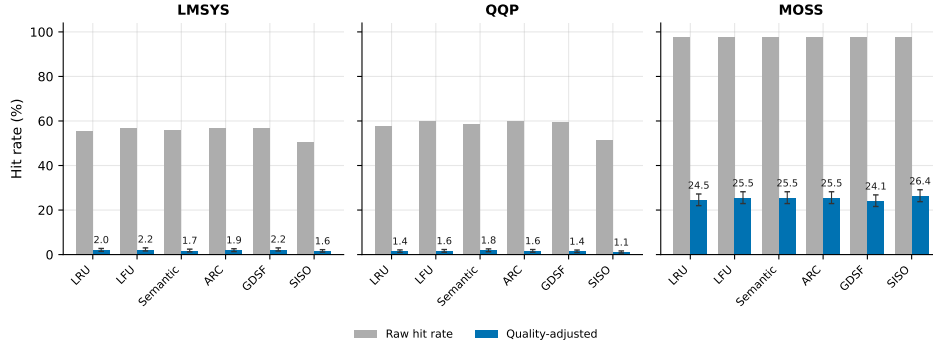


Figure 2: Raw and answer-substitutability-adjusted hit rate at 10% capacity under MiniLM (Wilson 95% intervals). The audited operating point admits few reusable LMSYS and QQP pairs.

other while both are resident. The claim does not cover arbitrary prefill contents, approximate-search false misses, insertion on hits, or policies that move stored representations.

Our density profile is consistent with this condition but does not prove it across the matrix. In the 10%-capacity MiniLM LRU cache, the mean within-threshold neighbor fraction at the end of the measured stream is 2.1×10^{-5} for QQP, 4.1×10^{-4} for LMSYS, and 2.5×10^{-3} for MOSS. The largest value is $40\times$ below the semantic policy’s smoothing constant $\mu = 0.1$, so redundancy contributes little to its ranking in this configuration.

5 Decision and validity limits

For an insert-on-miss cache under this protocol, LFU is the strongest simple default. Teams considering semantic eviction should first measure resident cache density at the intended hit radius. More importantly, they should set that radius against human-validated answer substitutability, not an embedding quantile alone.

Four limits constrain the ranking. Exact duplicate removal means the corpora do not measure repeated requests. The prefill protocol omits part of the first 30% and is not a continuous warmup. The common HNSW index controls the policy comparison, but its separately measured 2.2 pp static Recall@1 loss exceeds the differences among the leading policies; approximate misses also change later cache state, so exact-index replication is needed before interpreting sub-point ranks. Finally, the audit uses one 8B judge family, one encoder, one capacity, and six policies, without human labels or stored responses. Under these limits, LFU is preferable to the added policy complexity tested here; eviction is not universally irrelevant.

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A Judge rubric and validation

The judge is Llama-3.1-8B-Instruct served locally via Ollama at temperature 0 with a 16-token completion cap. The user prompt asks: “Would a correct, complete answer to the ORIGINAL question also be a correct and complete answer to the NEW question? Reply with exactly one word: YES or NO.” 13,667 of 13,669 pairs parse. This directional test measures answer substitutability, not symmetric query equivalence. QQP has no stored answers, so we could not perform a response-grounded audit. The 1,931-pair overlap between local Ollama and Groq’s hosted llama-3.1-8b-instant, using the same rubric, gives 98.55% agreement and $\kappa = 0.826$. Because both runtimes serve the same model family, this result measures consistency rather than correctness.

B Per-encoder recalibration

MiniLM’s hit threshold of 0.90 L_2^2 (cosine 0.55) is its median nearest-neighbor similarity over the 100K-query corpora. For gte-base, the hit and redundancy thresholds become 0.30 L_2^2 (cosine 0.85), the streaming-SISO clustering threshold moves from cosine 0.86 to 0.94, and ARC’s ghost-match threshold moves from 0.20 to 0.10 L_2^2 . Non-geometric parameters are unchanged. This in-sample distribution calibration makes the matrix non-degenerate; it is neither held-out calibration nor evidence that the resulting hits are valid.

C Reproducibility

Every run writes a JSON manifest with its command line, config hash, package versions, Git commit, and host hardware. Eviction cells use seeds {42, 123, 456}; the judge sample uses seed 42. The semantic policy sets $\alpha = \beta = 1.0$, $\mu = 0.1$, $\varepsilon = 10^{-9}$, and caps neighbor sampling at 1,024. At 10% capacity, per-query cost relative to LRU is $4.3\times$ (LMSYS) and $3.8\times$ (QQP) for the semantic policy, and $3.5\times$ and $3.1\times$ for streaming SISO.